



Fully woven display with smart sensors

An international team of scientists, led by researchers at the University of Cambridge, have produced a fully woven smart textile display that integrates active electronic sensing, energy, and photonic functions. The functions are em-



Researchers have developed a 116cm (46-inch) woven display with smart sensors, energy harvesting and storage integrated directly into the fabric (Credit: www.cam.ac.uk)

bedded directly into the fibres and yarns, which are manufactured using textile based industrial processes. Each fibre component is coated with materials that can withstand enough stretching. Some of the fibre based components are also braided to improve their reliability and durability. Finally, multiple fibre components are connected together using conductive adhesives and laser welding techniques. This approach could lead to applications such as curtains that work as TVs, energy-harvesting carpets, and interactive, self-powered clothing and fabrics.

Breakthrough for producing superior supercapacitor batteries

Existing supercapacitors are too large for many applications; they need to be about the same size as the battery they are connected to, which is an obstacle to integrating them in mobile phones or electric cars. Therefore, researchers at Chalmers University of Technology have developed a

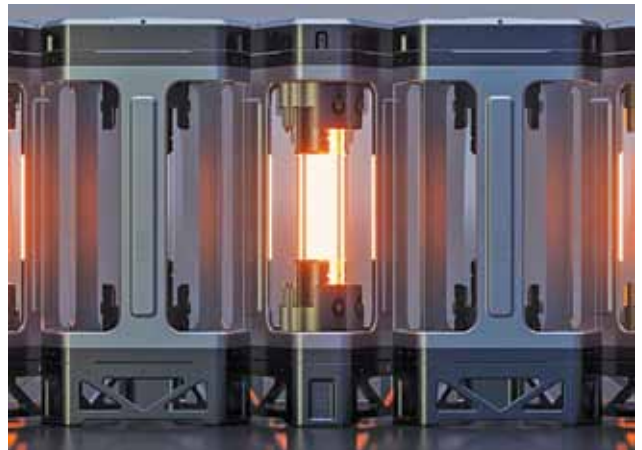


The image shows a 5cm wide silicon wafer with integrated micro supercapacitors, manufactured using the CMOS-compatible process (Credit: www.chalmers.se/en)

breakthrough method that can revolutionise how supercapacitors are produced. The proposed 'micro' supercapacitors are so small that they can fit on the system circuits that control various functions in mobile phones, computers, electric motors, and almost all electronics used today. Micro-supercapacitors of up to ten different materials can also be unified in a single manufacturing process, which means that properties can be easily tailored to suit several different end applications.

Possibility of having next-gen quantum batteries

In a key step towards making quantum batteries a reality, researchers at the University of Adelaide and their overseas partners have successfully proved the concept of superabsorption, a crucial idea underpinning quantum batteries.



Superabsorption unlocks key to next generation quantum batteries (Credit: www.adelaide.edu.au)

For this, the team built several wafer-like microcavities of different sizes which contained different numbers of organic molecules. Each was charged using a laser. As per a researcher, it is theoretically possible that the charging power of quantum batteries increases faster than the size of the battery, which could allow new ways of speed charging. The idea of the quantum battery has the potential to significantly impact energy capture and storage in renewable energy and in miniature electronic devices.

Thin sensor patches for measuring pulse waves

Pulse wave is an important bio-signal that indicates the status of cardiovascular health. Conventional pulse wave measuring devices, such as blood-pressure measuring cuff or stiff tong-shaped sensor, measure signals only at a single location, leading to less precise measurements. So, to ease